

Concavity, Curve Sketching, and Optimization



Increasing/decreasing and the first derivative test

- 1 For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.
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Concavity and the second derivative test

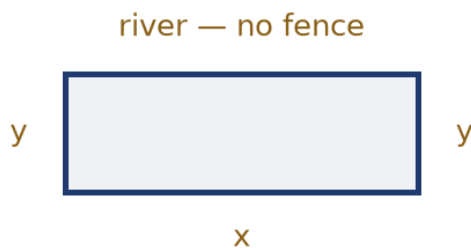
- 2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.
 - 3 Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 3x$.
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Curve sketching

- 4 For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.
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Optimization problems

- 5 A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.

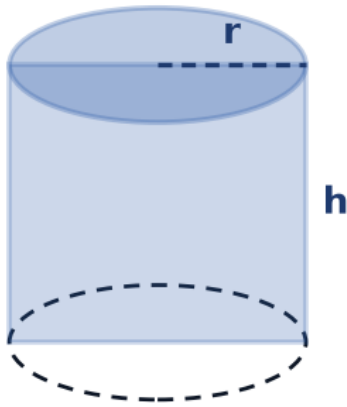


- 6 An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.
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Optimization: minimizing cost or distance

- 7 Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

- 8 A closed cylindrical can must have volume $16\pi \text{ cm}^3$. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi r h$, find the radius that minimizes surface area.

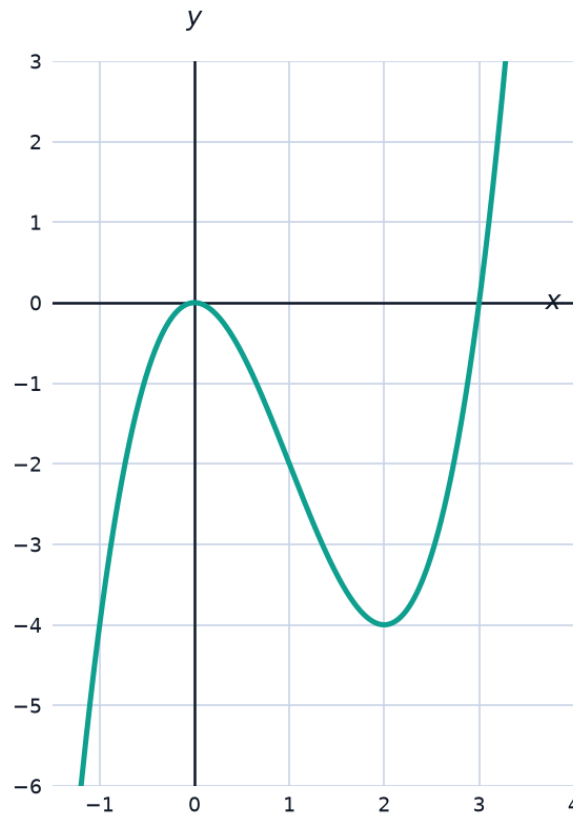


Points of inflection and graph behavior

- 9 Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

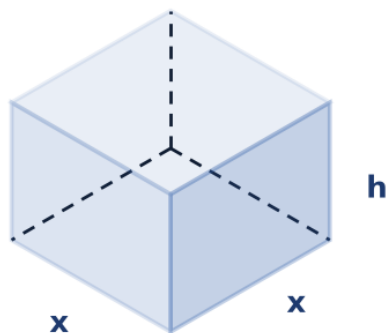
Visualizing extrema and concavity together

- 10 The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.



A capstone optimization problem

- 11 A rectangular box with a square base and no top must have volume 500 cm^3 . Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.



Second derivative test failures

- 12 For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.
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A capstone curve-sketching problem

- 13 For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.
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Multiple choice — easy 15

- 14 Classify the critical point $x = -1$ of $f(x) = x^3 + 6x^2 + 9x$, using the Second Derivative Test.
- a. a local maximum
 - b. cannot be determined without more information
 - c. a local minimum
 - d. an inflection point
- 15 Classify the critical point $x = -1$ of $f(x) = x^3 - 6x^2 - 15x$, using the Second Derivative Test.
- a. a local maximum
 - b. a local minimum
 - c. cannot be determined without more information
 - d. an inflection point
- 16 Classify the critical point $x = 1$ of $f(x) = x^3 + 3x^2 - 9x$, using the Second Derivative Test.
- a. cannot be determined without more information
 - b. a local minimum
 - c. an inflection point
 - d. a local maximum

- 17 Classify the critical point $x = 5$ of $f(x) = x^3 - 3x^2 - 45x$, using the Second Derivative Test.
- a. a local maximum
 - b. an inflection point
 - c. a local minimum
 - d. cannot be determined without more information
- 18 Classify the critical point $x = -6$ of $f(x) = x^3 + 12x^2 + 36x$, using the Second Derivative Test.
- a. a local minimum
 - b. a local maximum
 - c. an inflection point
 - d. cannot be determined without more information
- 19 Classify the critical point $x = 0$ of $f(x) = x^3 + 9x^2$, using the Second Derivative Test.
- a. cannot be determined without more information
 - b. a local minimum
 - c. a local maximum
 - d. an inflection point
- 20 Classify the critical point $x = 0$ of $f(x) = x^3 - 9x^2$, using the Second Derivative Test.
- a. a local maximum
 - b. an inflection point
 - c. cannot be determined without more information
 - d. a local minimum

- 21 Classify the critical point $x = 5$ of $f(x) = x^3 - 6x^2 - 15x$, using the Second Derivative Test.
- a. cannot be determined without more information
 - b. an inflection point
 - c. a local maximum
 - d. a local minimum
- 22 Classify the critical point $x = 6$ of $f(x) = x^3 - 3x^2 - 72x$, using the Second Derivative Test.
- a. cannot be determined without more information
 - b. an inflection point
 - c. a local maximum
 - d. a local minimum
- 23 Classify the critical point $x = 4$ of $f(x) = x^3 - 15x^2 + 72x$, using the Second Derivative Test.
- a. cannot be determined without more information
 - b. a local minimum
 - c. a local maximum
 - d. an inflection point
- 24 Classify the critical point $x = 4$ of $f(x) = x^3 - 9x^2 + 24x$, using the Second Derivative Test.
- a. a local maximum
 - b. an inflection point
 - c. a local minimum
 - d. cannot be determined without more information

- 25 Classify the critical point $x = 6$ of $f(x) = x^3 - 15x^2 + 72x$, using the Second Derivative Test.
- a. a local maximum
 - b. cannot be determined without more information
 - c. a local minimum
 - d. an inflection point
- 26 Classify the critical point $x = -6$ of $f(x) = x^3 + 9x^2$, using the Second Derivative Test.
- a. an inflection point
 - b. a local minimum
 - c. a local maximum
 - d. cannot be determined without more information
- 27 Classify the critical point $x = -2$ of $f(x) = x^3 + 3x^2$, using the Second Derivative Test.
- a. cannot be determined without more information
 - b. an inflection point
 - c. a local maximum
 - d. a local minimum
- 28 Classify the critical point $x = -5$ of $f(x) = x^3 + 3x^2 - 45x$, using the Second Derivative Test.
- a. an inflection point
 - b. a local maximum
 - c. cannot be determined without more information
 - d. a local minimum

Multiple choice — medium 15

29 Find the nonzero inflection point of $f(x) = x^4 + 6x^3$.

a. -3

b. -6

c. -2

d. 6

30 Find the nonzero inflection point of $f(x) = x^4 + 14x^3$.

a. 14

b. -6

c. -14

d. -7

31 Find the nonzero inflection point of $f(x) = x^4 + 10x^3$.

a. -4

b. -5

c. 10

d. -10

32 Find the nonzero inflection point of $f(x) = x^4 - 2x^3$.

a. -2

b. 2

c. 4

d. 1

33 Find the nonzero inflection point of $f(x) = x^4 + 4x^3$.

a. -1

b. -4

c. -2

d. 4

34 Find the nonzero inflection point of $f(x) = x^4 - 14x^3$.

a. 8

b. 14

c. 7

d. -14

35 Find the nonzero inflection point of $f(x) = x^4 + 12x^3$.

a. -5

b. -12

c. 12

d. -6

36 Find the nonzero inflection point of $f(x) = x^4 - 12x^3$.

a. 7

b. 6

c. -12

d. 12

37 Find the nonzero inflection point of $f(x) = x^4 + 16x^3$.

- a. -7
- b. -16
- c. -8
- d. 16

38 Find the nonzero inflection point of $f(x) = x^4 - 8x^3$.

- a. 5
- b. 8
- c. -8
- d. 4

39 Find the nonzero inflection point of $f(x) = x^4 - 4x^3$.

- a. 4
- b. -4
- c. 3
- d. 2

40 Find the nonzero inflection point of $f(x) = x^4 - 10x^3$.

- a. 5
- b. 6
- c. -10
- d. 10

41 Find the nonzero inflection point of $f(x) = x^4 + 2x^3$.

- a. 2
- b. -1
- c. -2
- d. -3

42 Find the nonzero inflection point of $f(x) = x^4 - 16x^3$.

- a. 16
- b. 8
- c. -16
- d. 9

43 Find the nonzero inflection point of $f(x) = x^4 + 8x^3$.

- a. -3
- b. -4
- c. 8
- d. -8

Multiple choice — hard 15

44 A farmer has 100 m of fencing to enclose a rectangular field with $x + 2y = 100$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

- a. 26
- b. 100
- c. 25
- d. 50

- 45 A farmer has 72 m of fencing to enclose a rectangular field with $x + 2y = 72$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.
- a. 72
 - b. 36
 - c. 19
 - d. 18
- 46 A farmer has 148 m of fencing to enclose a rectangular field with $x + 2y = 148$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.
- a. 37
 - b. 148
 - c. 74
 - d. 38
- 47 A farmer has 108 m of fencing to enclose a rectangular field with $x + 2y = 108$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.
- a. 28
 - b. 108
 - c. 27
 - d. 54
- 48 A farmer has 168 m of fencing to enclose a rectangular field with $x + 3y = 168$ (one side of length x needs no fence, the other 3 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.
- a. 168
 - b. 29
 - c. 28
 - d. 56

- 49 An open-top box is made from a 36 in by 36 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(36 - 2x)^2$.
- a. 6
 - b. 18
 - c. 9
 - d. 7
- 50 An open-top box is made from a 18 in by 18 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(18 - 2x)^2$.
- a. 3
 - b. 6
 - c. 9
 - d. 4
- 51 An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(12 - 2x)^2$.
- a. 4
 - b. 2
 - c. 3
 - d. 6
- 52 An open-top box is made from a 24 in by 24 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(24 - 2x)^2$.
- a. 4
 - b. 5
 - c. 12
 - d. 6

- 53 An open-top box is made from a 48 in by 48 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(48 - 2x)^2$.
- a. 9
 - b. 8
 - c. 24
 - d. 12
- 54 Find the x -coordinate of the point on the line $y = -5x - 4$ closest to the origin, by minimizing $D = x^2 + y^2$.
- a. $-\frac{4}{5}$
 - b. $\frac{2}{13}$
 - c. $-\frac{10}{13}$
 - d. $\frac{10}{13}$
- 55 Find the x -coordinate of the point on the line $y = 4x + 5$ closest to the origin, by minimizing $D = x^2 + y^2$.
- a. $-\frac{5}{17}$
 - b. $-\frac{5}{4}$
 - c. $-\frac{20}{17}$
 - d. $\frac{20}{17}$
- 56 Find the x -coordinate of the point on the line $y = 5x - 5$ closest to the origin, by minimizing $D = x^2 + y^2$.
- a. 1
 - b. $\frac{25}{26}$
 - c. $-\frac{25}{26}$
 - d. $\frac{5}{26}$

57 Find the x -coordinate of the point on the line $y = x + 3$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. $-\frac{3}{2}$

b. -3

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

58 Find the x -coordinate of the point on the line $y = x + 5$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. -5

b. $-\frac{5}{2}$

c. -3

d. $\frac{5}{2}$